

Mac Lee

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EDUCATION

Doctor of Philosophy in Physics University of California, San Diego, CA Advisor: Stefan G. Llewellyn Smith	October 2025
Master of Science in Physics with Distinction California State University, Northridge, CA Advisor: Donna D. N. Sheng	May 2018
Completed 24 units toward a Master of Music in Piano Performance University of Kansas, Lawrence, KS	Aug. 2011 – Dec. 2012
Bachelor of Music in Piano Performance Azusa Pacific University, Azusa, CA	July 2011

SCHOLARSHIPS/GRANTS/AWARDS/HONORS

Office of Naval Research Grant (UCSD)	2024
Election to Sigma Pi Sigma	2018
Mack I. Johnson Research Award for Outstanding Graduate Student in the College of Science and Mathematics (CSUN)	2018
C. Y. Liang Outstanding Graduate Student Award (CSUN)	2018
Leslie and Terry Cutler Scholarship (CSUN)	2017
Summer Research Grant (CSUN)	2016

TEACHING

Senior TA/Associate at UC San Diego	June 2022 – Sept. 2024
Lab Teaching Assistant Coordinator (LTAC/Head TA) of the PHYS 1-series labs	
Teaching Assistant at UC San Diego	Aug. 2018 – June 2022
Taught PHYS 1BL, 2BL, 2D, 10, and 140A. Lab Teaching Assistant Coordinator (LTAC/Head TA) of PHYS 2BL in Spring 2020, Spring 2021, Summer I 2021, and Spring 2022.	
Hathaway-Sycamores Child and Family Services Learning Center Instructor	Aug. 2013 – Sept. 2016

In my role as an instructor at Hathaway-Sycamores, I tutored K-12 students in language arts, math, and the sciences within the Los Angeles Unified School District (LAUSD). In addition to weekday tutoring sessions, I served as the math instructor for the Hathaway-Sycamores Summer SAT Workshop, which was organized in partnership with three local high schools from 2014 to 2016. I reprised this role as an SAT instructor in 2017 and 2018 as a volunteer.

Graduate Teaching Assistant at KU

Aug. 2011 – Aug. 2012

Taught PIAN 121, 221, 284, and 321

OPEN SOURCE CONTRIBUTIONS

FastVideo (open-source contributions)

May 2026 – present

Open-source inference and post-training framework for accelerated video generation; 3.5k+ GitHub stars, adopted/forked by SGLang for multimodal inference, integrated as a backend in NVIDIA Dynamo, and used by Tencent HY-WorldPlay. <https://github.com/hao-ai-lab/FastVideo>

- Collaborator / co-maintainer since June 2026
- Co-authored commits to port FLUX.2, a text-to-image flow-matching model, into FastVideo
- Ongoing work with a reinforcement learning training and inference pipeline, a performance benchmark regression tracking redesign, and Trajectory Distribution Matching, a distillation framework

TECHNICAL SKILLS

Programming Languages with proficiency: Python, Rust, OCaml, Haskell

Programming Languages with working knowledge: JavaScript, TypeScript, Julia, Lua, VimScript

Others: LLVM IR, MIPS Assembly, WebAssembly, HTML/CSS

PEER-REVIEWED PUBLICATIONS

- [4] M. Lee and S. G. Llewellyn Smith, “SQG point vortex dynamics with order rossby corrections”, [Eur. J. Mech. B Fluids](#) **118**, 204467 (2026).
- [3] M. Lee and S. G. Llewellyn Smith, “Stability of SQG kolmogorov flow”, [Dyn. Atmos. Oceans](#), 101680 (2026).
- [2] S.-S. Gong, W. Zheng, M. Lee, Y.-M. Lu, and D. N. Sheng, “Chiral spin liquid with spinon Fermi surfaces in the spin- $\frac{1}{2}$ triangular Heisenberg model”, [Phys. Rev. B](#) **100**, 241111 (2019).
- [1] M. Lee, T. R. Look, S. P. Lim, and D. N. Sheng, “Many-body localization in spin chain systems with quasiperiodic fields”, [Phys. Rev. B](#) **96**, 075146 (2017).

TALKS AND PRESENTATIONS

- [7] “SQG point vortex dynamics with order Rossby corrections”, SoCal Fluids XIX (University of California, Riverside, CA, Apr. 18, 2026), Talk.

- [6] “SQG point vortex dynamics with order Rossby corrections”, Workshop on Numerical and Theoretical Advances in Geophysical Fluid Dynamics and Nonlinear PDE Systems (Texas A&M University, College Station, TX, Feb. 7, 2026), Talk.
- [5] “Stability of SQG Kolmogorov flow with order Rossby corrections”, SoCal Fluids XVII (University of California, Irvine, CA, Apr. 13, 2024), Talk.
- [4] “Spin Liquid and Quantum Phase Diagram on a Spin-Orbit Coupled Triangular Lattice”, International Conference on Magnetism (San Francisco, CA, July 16, 2018), Poster Presentation.
- [3] “Spin Liquid and Quantum Phase Diagram on a Spin-Orbit Coupled Triangular Lattice”, APS March Meeting (Los Angeles, CA, Mar. 7, 2018), Talk.
- [2] “Topological Insulators”, Physics Symposium (California State University, Northridge, CA, July 2017), Poster Presentation.
- [1] “Many-Body Localization in Spin Chain Systems with Quasiperiodic Fields”, CSUNposium (California State University, Northridge, CA, Apr. 7, 2017), Talk.